	(32) $E = 3E_0 \cos\theta$	0,2	
2.10	(33) $\vec{p} = q\vec{a}$	0,4	**3,8**
	(34) $q = \rho\frac{4}{3}\pi r^3$	0,2	
	(35) $\vec{p} = 4\pi r^3 \varepsilon_0 \vec{E}$	0,4	
	(36) $\Delta E_z = \frac{\Delta q_i \cos\alpha}{4\pi\varepsilon_0 (z^2+R^2)^2}$	0,2	
	(37) $\cos\alpha = \frac{z}{(z^2+R^2)^{1/2}}$	0,2	
	(38) $E(z) = \frac{qz}{4\pi\varepsilon_0 (z^2+R^2)^{3/2}}$	0,4	
	(39) $F = qE(z+a) - qE(z) = qa\frac{dE}{dz} = p\frac{dE}{dz}$	0,4	
	(40) $F = \frac{q^2 r^3 z(R^2-2z^2)}{4\pi\varepsilon_0 (z^2+R^2)^4}$	0,2	
	(41) $z_1 = 0$	0,2	
	(42) $z_{2,3} = \pm\frac{R}{\sqrt{2}}$	0,2	
	(43) $F = -\frac{8q^2 r^3 x}{81\pi\varepsilon_0 R^6}$	0,4	
	(44) $x = z - \frac{R}{\sqrt{2}} \ll R.$	0,2	
	(45) $\ddot{x} + \frac{8q^2 r^3}{81\pi\varepsilon_0 R^6} x = 0$	0,2	
	(46) $\omega = \sqrt{\frac{8q^2 r^3}{81\pi\varepsilon_0 m R^6}}$	0,2	
2.11	(47) $A = 0$	0.4	**0,4**
	(40)	(0.1)	
합계			**10,0**

문제 3. 레이저 (10.0점)

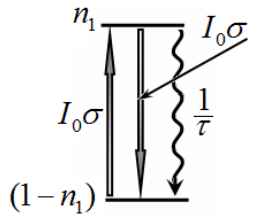
분포 반전: 2준위계

3.1 .

n_1 ,

$(1-n_1)$.

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$$\frac{dn_1}{dt} = -\frac{1}{\tau}n_1 - I_0\sigma n_1 + I_0\sigma(1-n_1). \quad (1)$$

3.2 $dn_1/dt = 0$, (1)

$$\bar{n}_1 = \frac{I_0\sigma\tau}{1+2I_0\sigma\tau}. \quad (2)$$

$$\Delta\bar{n} = \bar{n}_1 - (1-\bar{n}_1) = 2\frac{I_0\sigma\tau}{1+2I_0\sigma\tau} - 1 = -\frac{1}{1+2I_0\sigma\tau}. \quad (3)$$

3.3 2

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